Haithem Turki

Curriculum Vitae

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Education

Carnegie Mellon University **2019 - 2024**
Ph.D. in Computer Science
Thesis: Towards City-Scale Neural Rendering
Advisor: Deva Ramanan (committee chair)
Committee: Jessica K. Hodgins, Martial Hebert, Jonathan T. Barron

Stanford University **2012 - 2018**
M.S. in Computer Science

Stanford University **2008 - 2012**
B.S. in Computer Science, minor in Management Science and Engineering

Experience

NVIDIA, Senior Research Scientist **2024-**
Spatial Intelligence Lab with Zan Gojcic and Sanja Fidler

Meta, Research Scientist Intern w/ Christian Richardt, Michael Zollhöfer **2023**
Argo AI, Research Intern w/ Deva Ramanan, Francesco Ferroni **2022**
Page One Ventures, Venture Partner **2020 - 2024**
Palantir Technologies, Engineering Manager and Technical Lead **2012 - 2019**
Addepar, Engineering Intern **2011**
Stanford University, Research Assistant w/ Daphne Koller, M. Pawan Kumar **2010**

Awards & Honors

Best Paper Award Candidate, CVPR 2025 **2025**
DIFIX3D+: Improving 3D Reconstructions with Single-Step Diffusion Models

Highlight Paper, CVPR 2024 (top 2.8% of submissions) **2024**
HybridNeRF: Efficient Neural Rendering via Adaptive Volumetric Surfaces

Highlight Paper, CVPR 2024 (top 2.8% of submissions) **2024**
SpecNeRF: Gaussian Directional Encoding for Specular Reflections

Publications

[18] Alessandro Burzio*, Tobias Fischer*, Sven Elflein, Qunjie Zhou, Riccardo de Lutio, Jiawei Ren, Jiahui Huang, Shengyu Huang, Marc Pollefeys, Laura Leal-Taixé, Zan Gojcic†, **Haithem Turki**†,“Déjà View: Looping Transformers for Multi-View 3D Reconstruction”, Preprint

[17] Riccardo de Lutio*, Tobias Fischer, Yen-Yu Chang, Yuxuan Zhang, Jay Zhangjie Wu, Xuanchi Ren, Tianchang Shen, Katarina Tothova, Zan Gojcic, **Haithem Turki***,“ArtiFixer: Enhancing and Extending 3D Reconstruction with Auto-Regressive Diffusion Models” in ACM SIGGRAPH, 2026

[16] Yuxuan Zhang*, Katarina Tothova*, Zian Wang, Kangxue Yin, **Haithem Turki**, Riccardo de Lutio, Yen-Yu Chang, Or Litany, Sanja Fidler, Zan Gojcic,“DiffusionHarmonizer: Bridging Neural Reconstruction and Photorealistic Simulation with Online Diffusion Enhancer” in IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2026

[15] **Haithem Turki***, Qi Wu*, Xin Kang, Janick Martinez Esturo, Shengyu Huang, Ruilong Li, Zan Gojcic, Riccardo de Lutio,“SimULi: Real-Time LiDAR and Camera Simulation with Unscented Transforms” in International Conference on Learning Representations (ICLR), 2026

[14] Sherwin Bahmani, Tianchang Shen, Jiawei Ren, Jiahui Huang, Yifeng Jiang, **Haithem Turki**, Andrea Tagliasacchi, David B. Lindell, Zan Gojcic, Sanja Fidler, Huan Ling, Jun Gao*, Xuanchi Ren*,“Lyra: Generative 3D Scene Reconstruction via Video Diffusion Model Self-Distillation” in International Conference on Learning Representations (ICLR), 2026

[13] Zhangjie Wu*, Yuxuan Zhang*, **Haithem Turki**, Xuanchi Ren, Jun Gao, Mike Zheng Shou, Sanja Fidler, Zan Gojcic†, Huan Ling†,“DIFIX3D+: Improving 3D Reconstructions with Single-Step Diffusion Models” in IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2025, ***(Best Paper Award Candidate)***

[12] **Haithem Turki**, Vasu Agrawal, Samuel Rota Bulò, Lorenzo Porzi, Peter Kontschieder, Deva Ramanan, Michael Zollhöfer, Christian Richardt,“HybridNeRF: Efficient Neural Rendering via Adaptive Volumetric Surfaces” in IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2024, ***(Highlight)***

[11] Li Ma, Vasu Agrawal, **Haithem Turki**, Changil Kim, Chen Gao, Pedro Sander, Michael Zollhöfer, Christian Richardt,“SpecNeRF: Gaussian Directional Encoding for Specular Reflections” in IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2024, ***(Highlight)***

[10] **Haithem Turki**, Michael Zollhoefer, Christian Richardt, Deva Ramanan,“PyNeRF: Pyramidal Neural Radiance Fields” in Conference on Neural Information Processing Systems (NeurIPS), 2023

[9] Shilpa George, **Haithem Turki**, Ziqiang Feng, Thomas Eiszler, Deva Ramanan, Padmanabhan Pillai, Mahadev Satyanarayanan,“Low-Bandwidth Self-Improving Transmission of Rare Training Data” in Annual International Conference on Mobile Computing And Networking (MobiCom), 2023

[8] Shilpa George, **Haithem Turki**, Ziqiang Feng, Thomas Eiszler, Deva Ramanan, Padmanabhan Pillai, Mahadev Satyanarayanan,“Edge-based Privacy-Sensitive Live Learning for Discovery of Training Data” in International Workshop on Networked AI Systems (NetAISys), 2023

[7] **Haithem Turki**, Jason Y. Zhang, Francesco Ferroni, Deva Ramanan,“SUDS: Scalable Urban Dynamic Scenes” in IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2023

[6] Mahadev Satyanarayanan, Ziqiang Feng, Shilpa George, Jan Harkes, Roger Iyengar, **Haithem Turki**, Padmanabhan Pillai,“Accelerating Silent Witness Storage” in IEEE Micro, Nov.-Dec. 2022

[5] **Haithem Turki**, Deva Ramanan, Mahadev Satyanarayanan,“Mega-NeRF: Scalable Construction of Large-Scale NeRFs for Virtual Fly-Throughs” in IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2022

[4] Shilpa George, Thomas Eiszler, Roger Iyengar, **Haithem Turki**, Ziqiang Feng, Junjue Wang, Padmanabhan Pillai, Mahadev Satyanarayanan,“OpenRTiST: End-to-End Benchmarking for Edge Computing” in IEEE Pervasive Computing, vol. 19, no. 4, pp. 10-18, Oct.-Dec. 2020

[3] Mahadev Satyanarayanan, Thomas Eiszler, Jan Harkes, **Haithem Turki** and Ziqiang Feng,“Edge Computing for Legacy Applications” in IEEE Pervasive Computing, Oct.-Dec. 2020

[2] M. Pawan Kumar, **Haithem Turki**, Dan Preston and Daphne Koller,“Parameter Estimation and Energy Minimization for Region-Based Semantic Segmentation” in IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 37, no. 7, pp. 1373-1386, July 2015

[1] M. Pawan Kumar, **Haithem Turki**, Dan Preston and Daphne Koller,“Learning Specific-Class Segmentation from Diverse Data” in International Conference on Computer Vision (ICCV), 2011

Technical Reports

[2] Tianshi Cao*, Jiawei Ren*, Yuxuan Zhang, Jaewoo Seo, Jiahui Huang, Shikhar Solanki, Haotian Zhang, Mingfei Guo, **Haithem Turki**, Muxingzi Li, Yue Zhu, Sipeng Zhang, Zan Gojcic, Sanja Fidler, Kangxue Yin,“Asset Harvester: Extracting 3D Assets from Autonomous Driving Logs for Simulation”, arXiv:2604.18468, 2026

[1] Ziqiang Feng, Shilpa George, **Haithem Turki**, Roger Iyengar, Padmanabhan Pillai, Jan Harkes, Mahadev Satyanarayanan,“Improving Edge Elasticity via Decode Offload”, Technical Report CMU-CS-21-139, Carnegie Mellon University, 2021

Patents

[2] **Haithem Turki**, Robert Fink, Amr Al Mallah,“Systems and Methods for Indexing and Searching.” US Patent 2019/0347343 A1, 2019

[1] **Haithem Turki**, Sander Kromwijk, Stephen Cohen, Yixun Xu, Feridun Arda Kara,“Systems and Methods for Constraint Driven Database Searching.” US Patent 2018/293239 A1, 2018

Teaching

15-640 Distributed Systems, Carnegie Mellon University (Graduate Student Instructor) — 2020 - 2021

17-700 Data Science and Machine Learning at Scale, Carnegie Mellon University (Graduate Student Instructor) — 2020 - 2021

15-821 Mobile and Pervasive Computing, Carnegie Mellon University (Project Mentor) — 2020 - 2021

Student Collaborators & Interns

Tobias Fischer, PhD Student @ ETH Zürich. See Publications [17, 18].

Alessandro Burzio, PhD Student @ University of Modena and Reggio Emilia. See Publications [18].

Invited Talks

"Towards City-Scale Neural Rendering"

Stanford Vision Lab	2024
Waymo Research (virtual)	2024
Google (virtual)	2024
NVIDIA Toronto AI Lab (virtual)	2024

"Mega-NeRF: Scalable Construction of Large-Scale NeRFs for Virtual Fly-Throughs"

Argo AI	2022

Academic Service

Area Chair:
ACCV

Reviewer:
CVPR, ACCV, ECCV, ICCV, NeurIPS, ICLR, SIGGRAPH, SIGGRAPH Asia, AAAI, Transactions on Image Processing

Additional Information

Languages:
English (native), French (fluent), Arabic (fluent), Japanese (elementary), German (elementary)

Other Qualifications:
Alchemist Accelerator (startup mentor and investor). FINRA Series 65. Graduate of the Stanford StartX startup accelerator.